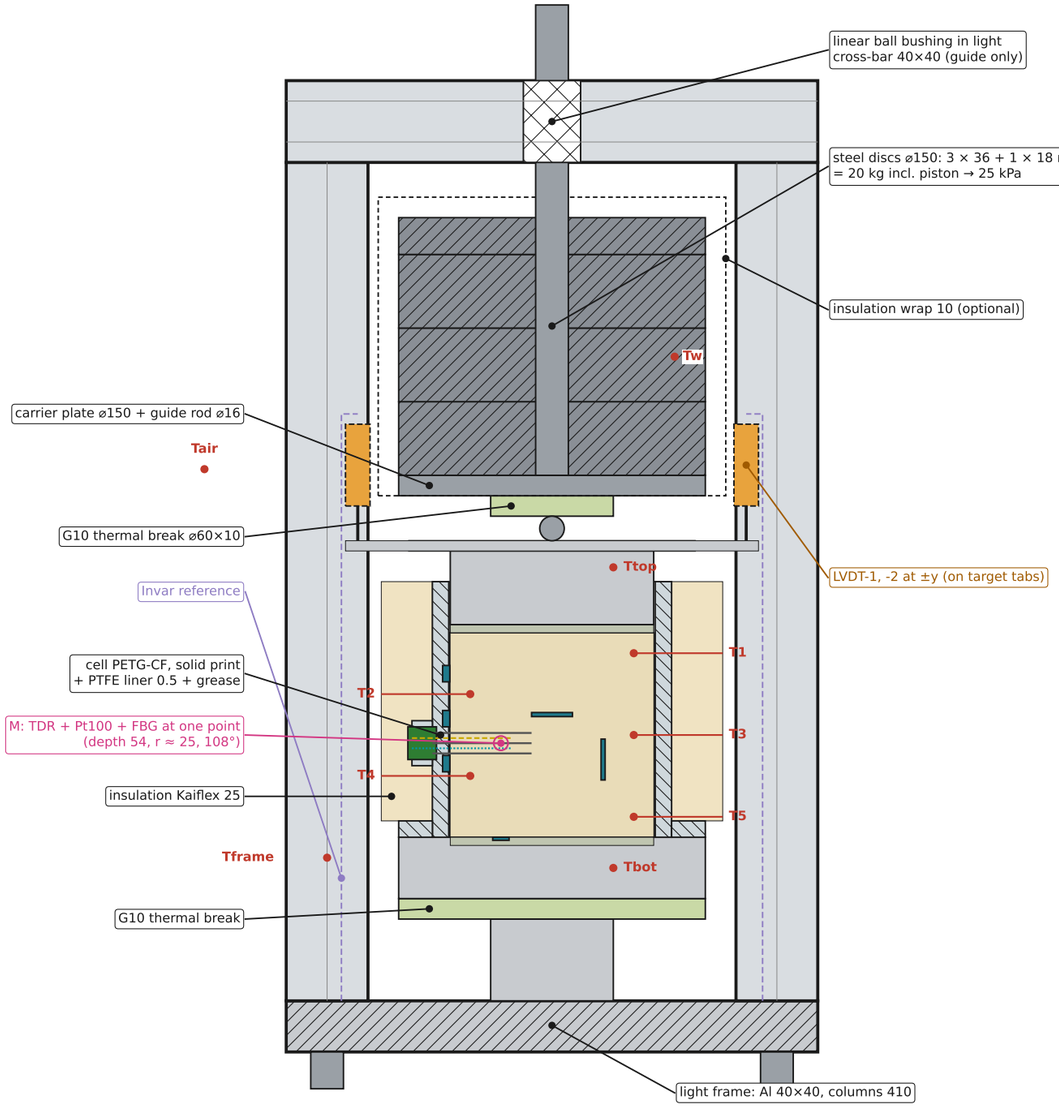


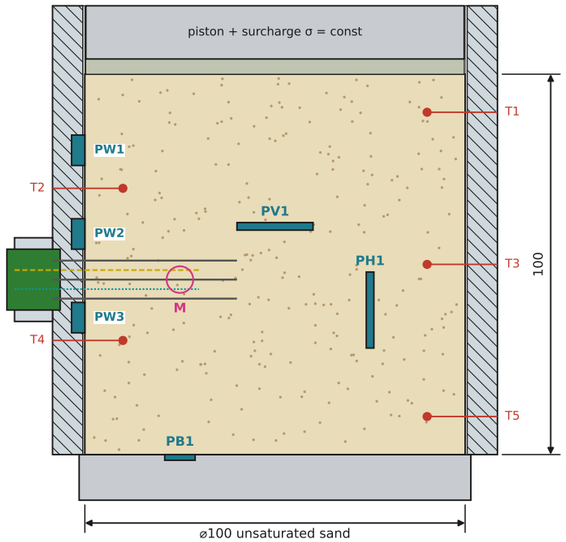
Sensors (tag → model → CR1000)

tag	measures	recommended model	CR1000
PV1 / PH1	vert. / radial stress	TML PDA-PA 200 kPa on ø20×2 disc	MUX-1 (4×16), BrFull, VX3 2 V
PB1	vertical stress at base	TML PDA-PA 200 kPa (flush)	MUX-1 (4×16), BrFull, VX3 2 V
PW1…PW3	lateral stress on wall	TML PDA-PA 200 kPa (flush)	MUX-1 (4×16), BrFull, VX3 2 V
LVDT-1,2	check: no movement	RDP DCTH400	SE1–2, VoltSE ±5 V
T1…T5	soil, r 40 mm	Campbell 109SS	MUX-2 → SE13, VX2
M: TDR	water / unfrozen water	mini 3-rod TDR 40 mm [U] + TDR200	SDM C1–C3
M: Pt100	reference temperature	Pt100 1/10 DIN, 4-wire, ø3	MUX-1 + 4WPB100, BrHalf4W
M: FBG	optical temperature	FBG temp. sensor in capillary ø1 [U]	PC (Ethernet), time-synced
Ttop, Tbot, Tw	piston, pedestal, weights	Campbell 109SS	(same)
Tframe, Tair	frame, air	Campbell 109SS / 107	MUX-2 → SE13, VX2

σ kPa	total kg	full discs	half discs	trim kg
5	4.0	0	0	1.25
10	8.0	1	0	0.33
15	12.0	1	1	1.88
20	16.0	2	1	0.96
25	20.0	3	1	0.04

How it works
Gravity gives an exactly constant surcharge; fixed part ≈ 2.8 kg ≈ 3.4 kPa (piston, carrier, rod).
No load cell, no adjustment; the frame only guides the rod and carries the LVDTs.
Weights lag the freezer air by ≈ 0.6 K at the 0.1 K/h ramp (vypocty.md §7).

DETAIL – sand, pressure sensors, thermistors ≈ 1 : 1.6



schematic; orange = displacement, blue = pressure, red = temperature; cells and LVDTs revolved into section

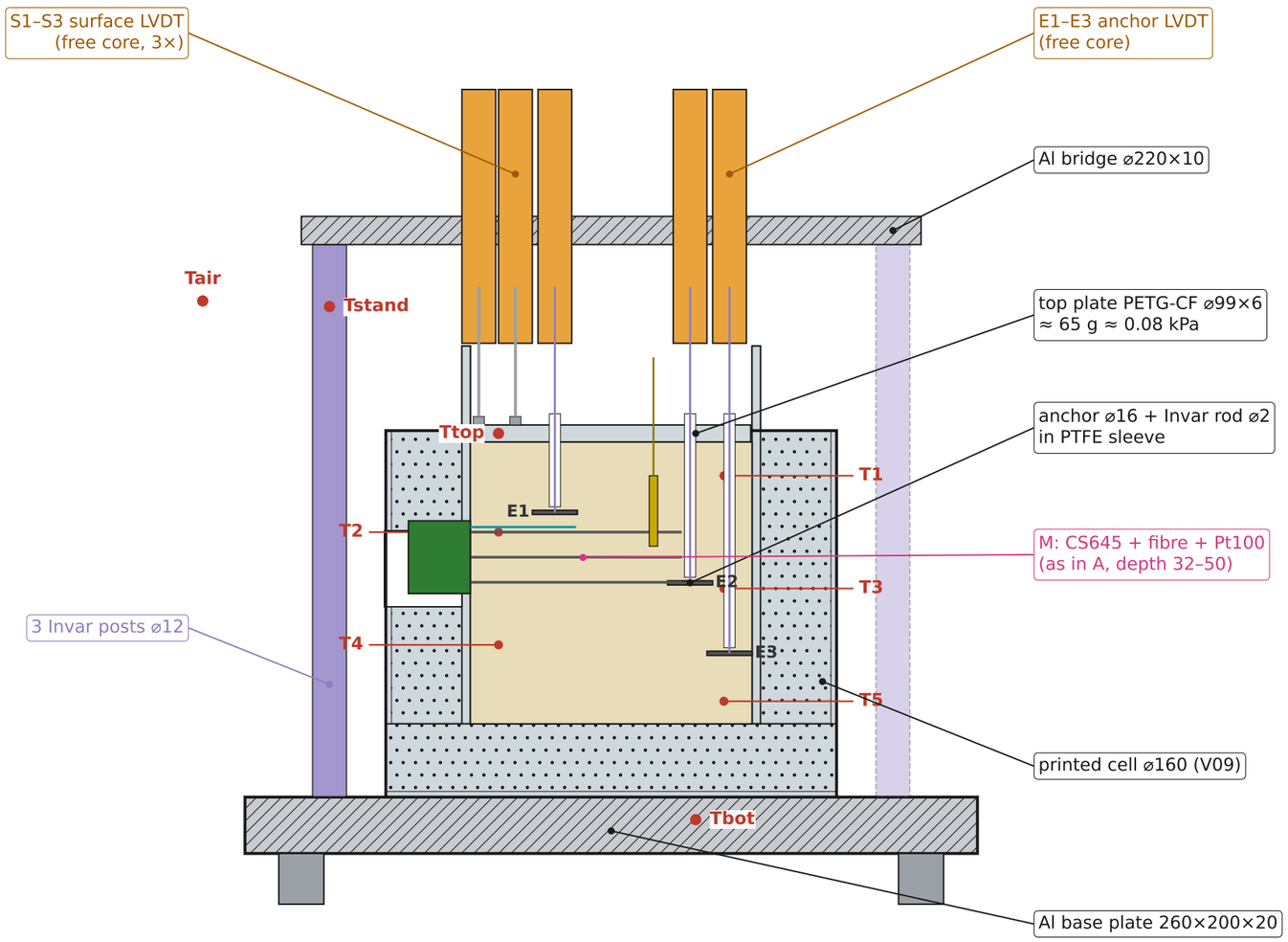
Variant A – unsaturated sand, dead-weight surcharge ≤ 25 kPa

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**≈ 1 : 3**	drawing no.	**E7-V01**	sheet 1 / 10

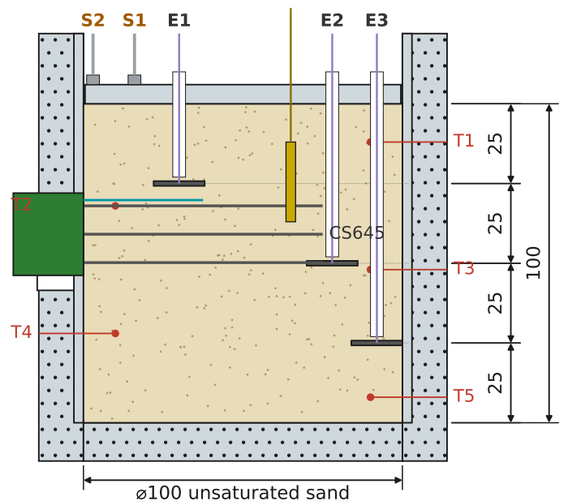
Sensors (tag → model → CR1000)

tag	measures	model	CR1000
S1…S3	surface heave	RDP DCTH200, free core [U]	SE3–5
E1…E3	anchors 25/50/75	RDP DCTH200, free core [U]	SE6–8
M: TDR	unfrozen water	Campbell CS645 (existing)	C1–C3 SDM
M: Pt100	temperature	Pt100 37 mm (existing)	4-wire
M: OF	temperature	optical fibre (existing)	own reader
T1…T5	soil, r 40	Campbell 109SS	MUX-2 SE13
Ttop, Tbot	plate, base	Campbell 109SS	(same)
Tstand, Tair	stand, air	Campbell 109SS / 107	(same)

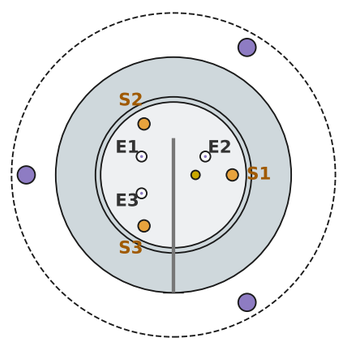
Per freeze–thaw cycle
• heave, settlement, residual: mean of S1–S3 (difference = tilt)
• 4 layers of 25 mm: $\bar{S}$ − E1, E1 − E2, E2 − E3, E3 (bottom fixed)
  → strain per layer, resolution ≈ $4 \cdot 10^{-5}$
• frost depth from T1–T5, unfrozen water at M
No load: free-core LVDTs push nothing, the plate adds 0.08 kPa.
Reference error ≈ 40 µm per 20 K → blank run (vypocty.md §4).



DETAIL – anchors ≈ 1 : 1.9



PLAN 1 : 4

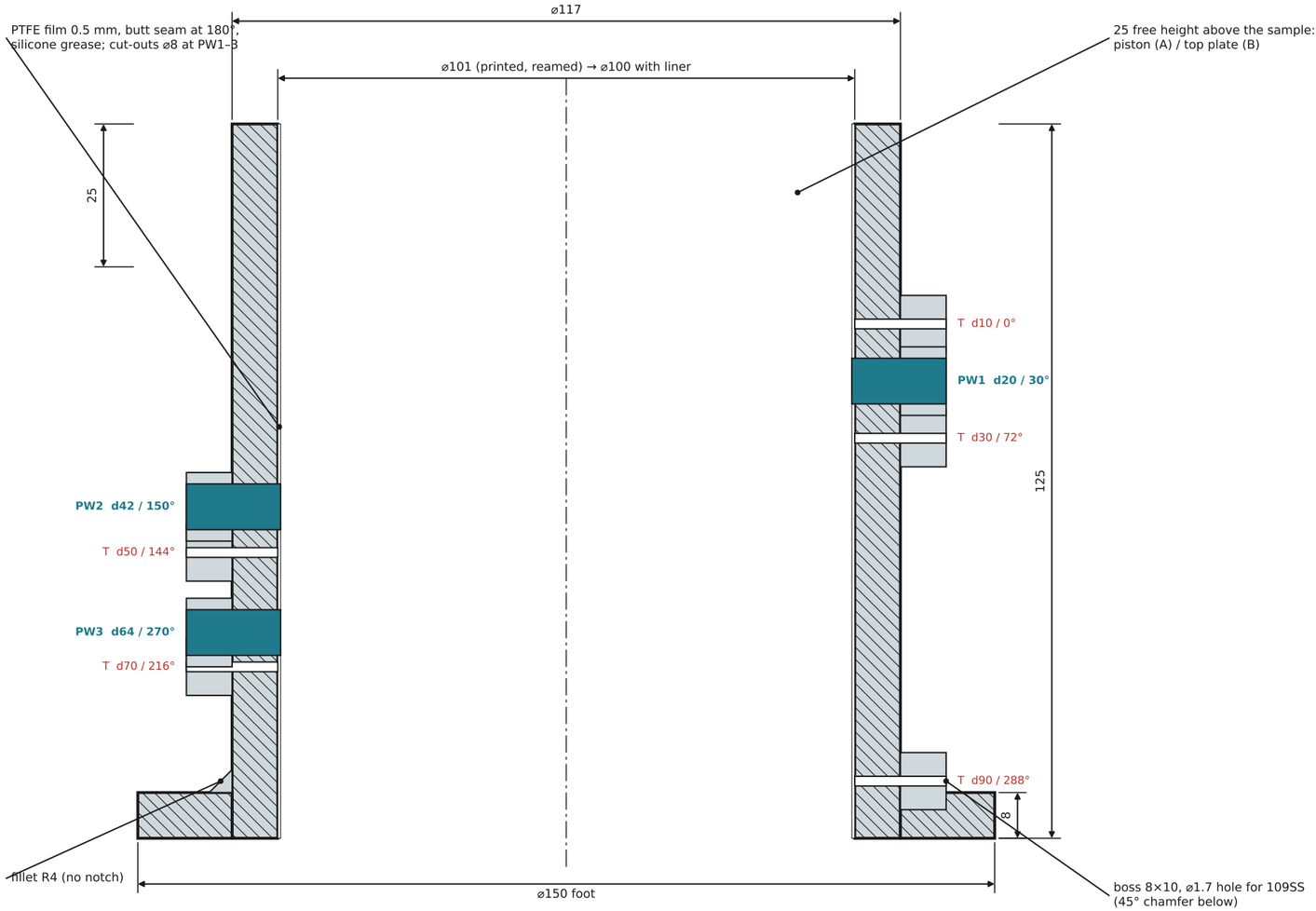


S = surface LVDT (r 40)
E = anchor (r 25)
yellow = Pt100, green = port
violet = Invar posts

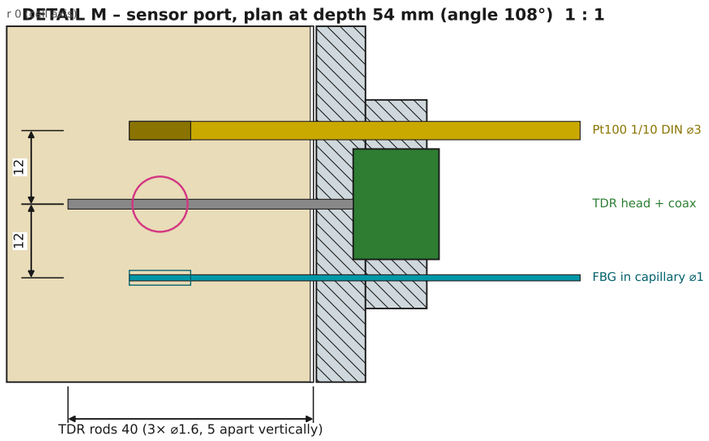
orange = displacement, red = temperature, violet = Invar; revolved section

Variant B – same sample, no load: heave and layer strain

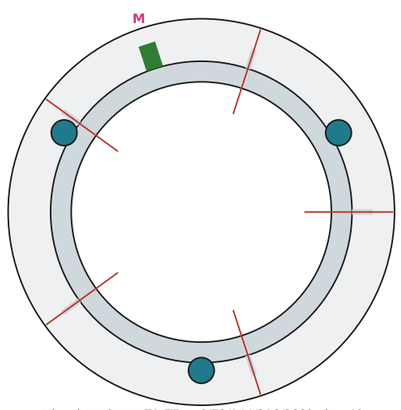
status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**≈ 1 : 2**	drawing no.	**E7-V02**	sheet 2 / 10



VIEW F – cell from above 1 : 2.4



DETAIL M – sensor port, plan at depth 54 mm (angle 108°) 1 : 1



red = thermistors T1–T5 at 0/72/144/216/288°, tip r 40;
blue = PW1–3 at 30/150/270°; green = point M at 108°

Material check – PETG-CF

load case	stress	SF short	SF long
hoop: A 25 kPa surcharge, K0 = 1	0.16 MPa	> 250	> 75
hoop: B ice on the wall 200 kPa (upper est.)	1.3 MPa	32–44	9–13
hoop: extreme 1 MPa	6.3 MPa	6–9	1.9–2.6
axial (across layers): friction, A 25 kPa	< 0.05 MPa	> 300	> 90

Assumed strength (data-sheet range, verify): XY 40–55 MPa, Z 15–30 MPa, E 3–6 GPa; long-term limit ≈ 30 % (creep, layer bond). Details: vypocty.md §5.

Verdict: suitable for both variants as a solid print (no sleeve needed).

Wall friction (the price of a solid cell, vypocty.md §5)
- A: friction carries part of the surcharge into the wall; with the PTFE liner + grease (μ ≈ 0.05) the vertical stress at the base is ≈ 90 % of the top, without the liner (μ ≈ 0.5) only ≈ 30–45 % → compare PV1 / PB1 with it.
- B: friction and ice adhesion hold back heave of the frozen top layer; PTFE has very low ice adhesion – liner and grease are mandatory.

Conditions
- wall 8 mm, concentric perimeters only, 100 % infill;
- no printed threads or sharp notches (CF lowers toughness at −10 °C);
- inside sealed with a thin epoxy coat before the liner goes in;
- cell not clamped at the foot – friction then pushes, never pulls layers.

Print settings (starting point)
hardened nozzle 0.6 mm, layer 0.2 mm, 250–260 °C, bed 80 °C, slow outer walls, filament dried 4 h at 65 °C, enclosure, part cooling ≤ 30 %.
Print upright, bosses with 45° chamfer underneath (no supports).

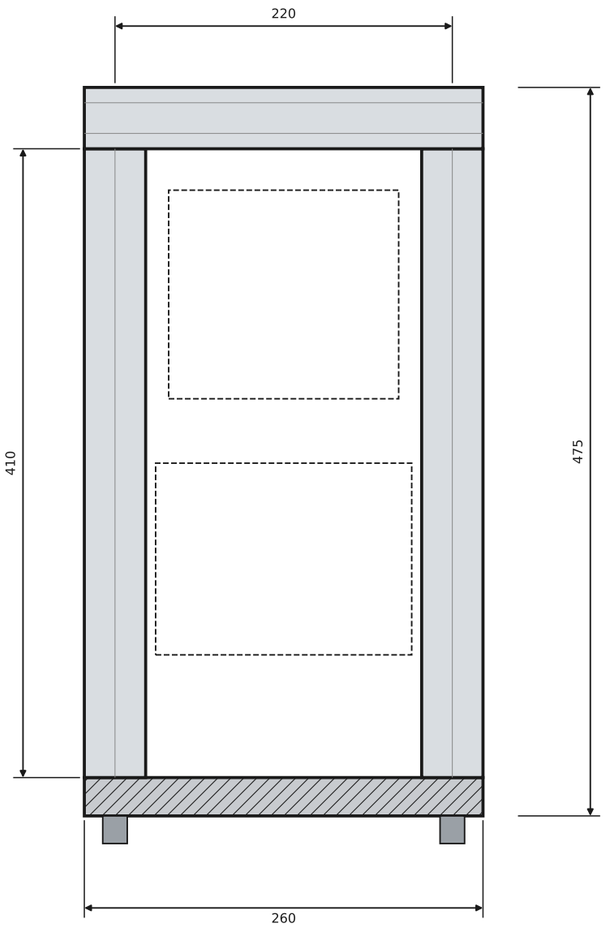
Qualification before use
1 10 freeze–thaw cycles −10/+10 °C of water-saturated coupons, then tensile test XY and Z vs. dry control;
2 bore check ⌀101 ± 0.1 at 3 heights, at +20 and −10 °C.

print upright in one piece; hoop direction = layer direction; foot rests on the pedestal, not bolted

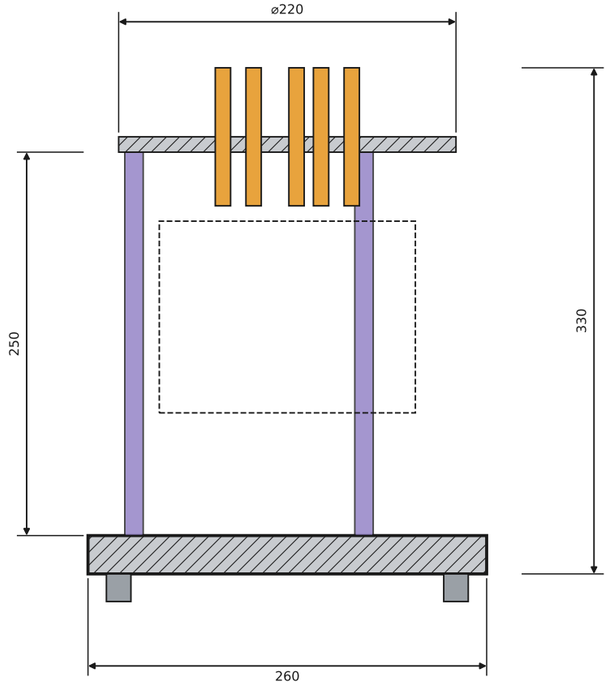
Sample cell – PETG-CF, solid 3D print (both variants)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 1**	drawing no.	**E7-V03**	sheet 3 / 10

A – FRONT VIEW 1 : 4



B – FRONT VIEW 1 : 4



Bill of materials

#	part	size	qty
1	base plate Al 6082 T6 (A and B)	260 × 200 × 25	2
2	A: profile 40×40 column	l = 410	2
3	A: profile 40×40 cross-bar	l = 260	1
4	A: linear ball bushing ⌀16 in cross-bar		1
5	A: Invar post ⌀10 + clamp arm (LVDT)	l ≈ 300	2
6	B: Invar post ⌀12, M6 both ends	l = 250	3
7	B: Al bridge ⌀220×10, 6 LVDT clamps		1
8	core bolt M8 / socket screw M8		8
9	levelling foot M12, steel		6

Frame A: carries nothing except the guide of the weight rod (bushing)
and the LVDT posts; the surcharge goes straight down through the sample.

Stand B: three Invar posts screwed into the base plate at r 100
(60/180/300°, outside the insulation), bridge on top with split clamps
for 5 LVDTs (+1 spare). The bridge is lifted off to fill the cell.
Reference error of the stand ≈ 50 µm per 20 K (Al pedestal vs. Invar)
→ blank run with an Al dummy sample (vypocty.md §4).

Heights incl. base plate: A 475 mm (+ guide rod), B 330 mm;
two rigs side by side on the tray – layout on E7-V06.

frame A only guides the weights; stand B is only a displacement reference – neither carries load

Frame A (aluminium profiles) and Invar stand B

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 4**	drawing no.	**E7-V04**	sheet 4 / 10

Options per measurement (PICK = recommended)

sensor	maker / model	key specs	verdict
LVDT ±10	RDP DCTH400	±5 V / 0–10 V out, −40…+80 °C, ±0.25 % FS	PICK (A)
LVDT ±10	Micro-Epsilon DTA-10D + MSC7401	0–5 V / 4–20 mA, −20…+80 °C	alternative
LVDT ±10	Solartron AX/10/S + OD5	±5 V out, needs 230 V conditioner	alternative
LVDT ±10	HBM WA-L 20 mm	passive, needs carrier amplifier	alternative
LVDT free	RDP DCTH200 unguided [U]	±5 mm, no spring, core on ø3 rod; rig B	PICK (B)
LVDT free	Micro-Epsilon DTA-5G8 + MSC7401 [U]	±5 mm, free core, 0–5 V, −20…+80 °C	alternative
LVDT free	Solartron AX/5/F (free core) + OD5 [U]	±5 mm, ±5 V via conditioner	alternative
laser	Micro-Epsilon optoNCDT 1420	non-contact, but op. 0…+50 °C, frost on window	not suitable
LVDT ±10	TE DC-EC 500	operating only 0…+70 °C	not suitable
p cell	TML PDA-PA 200 kPa	ø6.5×1 mm, 0.8 mV/V, comp. −10…+60, op. −20…+70 °C	PICK
p cell	Kyowa BEC-A 200/500 kPa	ø30 mm, for model tests; t, temp. [U]	one cell only
p cell	TML KDE/KDF-PA	ø50 mm – too large for ø100 sample	not suitable
p wall	TE/MEAS EPL	miniature, comp. −40…+90 °C, ø [U]	alternative
p cell	Tekscan FlexiForce A201	thin film, drift < 5 %/log t	comparison only
T probe	Campbell 109SS	ø1.6 mm, ±0.38 °C at 0 °C, direct on CR1000	PICK
T probe	Campbell 107	ø7.6 mm, ±0.2 °C (0–50 °C)	air / frame
T probe	TE/YSI 44031 bead	±0.1 °C, own half bridge needed	for 0.01 °C (E6 chain)
Pt100	Pt100 1/10 DIN 4-wire ø3 + 4WPB100	±0.03 °C at 0 °C, via MUX-1	PICK (point M)
TDR	mini 3-rod 40 mm (custom) + TDR200	fits ø100 sample; calibrate in the sand	PICK [U]
TDR	Campbell CS645 (3 rods 75 mm)	too long for r 50 – would cross the axis	not suitable
TDR	Easy Test LP/ms micro probe + own meter [U]	rods ≈ 53 mm; separate logger	alternative
optical	FBG in capillary ø1 + HBM FS22 / Luna si155 [U]	≈ ±0.2–0.5 °C, many points on one fibre	PICK [U]
optical	fluorescent probe Opsens OTG-F + TempSens [U]	±0.3 °C, analog out → CR1000 SE	alternative

CR1000 channel plan

terminal	connected to	measurement
SE1–SE2	LVDT-A1, A2 (rig A)	VoltSE ±5000 mV, averaging; own ±15 V supply
SE3–SE5	LVDT S1–S3 (rig B surface)	VoltSE ±5000 mV
SE6–SE8	LVDT E1–E3 (rig B anchors)	VoltSE ±5000 mV
DIFF5, DIFF6	MUX-1 AM16/32B, 4×16	9 p-bridges + 2 Pt100 (4WPB100); COM H2/L2 → DIFF5
VX3	MUX-1 excitation	2000 mV bridges / 2500 mV Pt100
C1–C3	TDR200 (SDM) → SDMX50SP	2 TDR probes (point M, A and B)
C4, C5	MUX-1 reset, clock	
SE13	MUX-2 AM16/32B, 2×32	≈ 20 thermistors (Therm109), VX2
C6, C7	MUX-2 reset, clock	
12V, G	both multiplexers, TDR200	
–	FBG interrogator	1 fibre for both rigs; own PC, clock synced
SE14–16	spare	

Channel count (2 rigs): 8 LVDTs, 9 pressure bridges, ≈ 20 thermistors,
point M ×2: 2 TDR (TDR200 + SDMX50SP), 2 Pt100 (MUX-1), 2 FBG (interrogator).
No load cells: A is loaded by dead weight, B is not loaded at all.

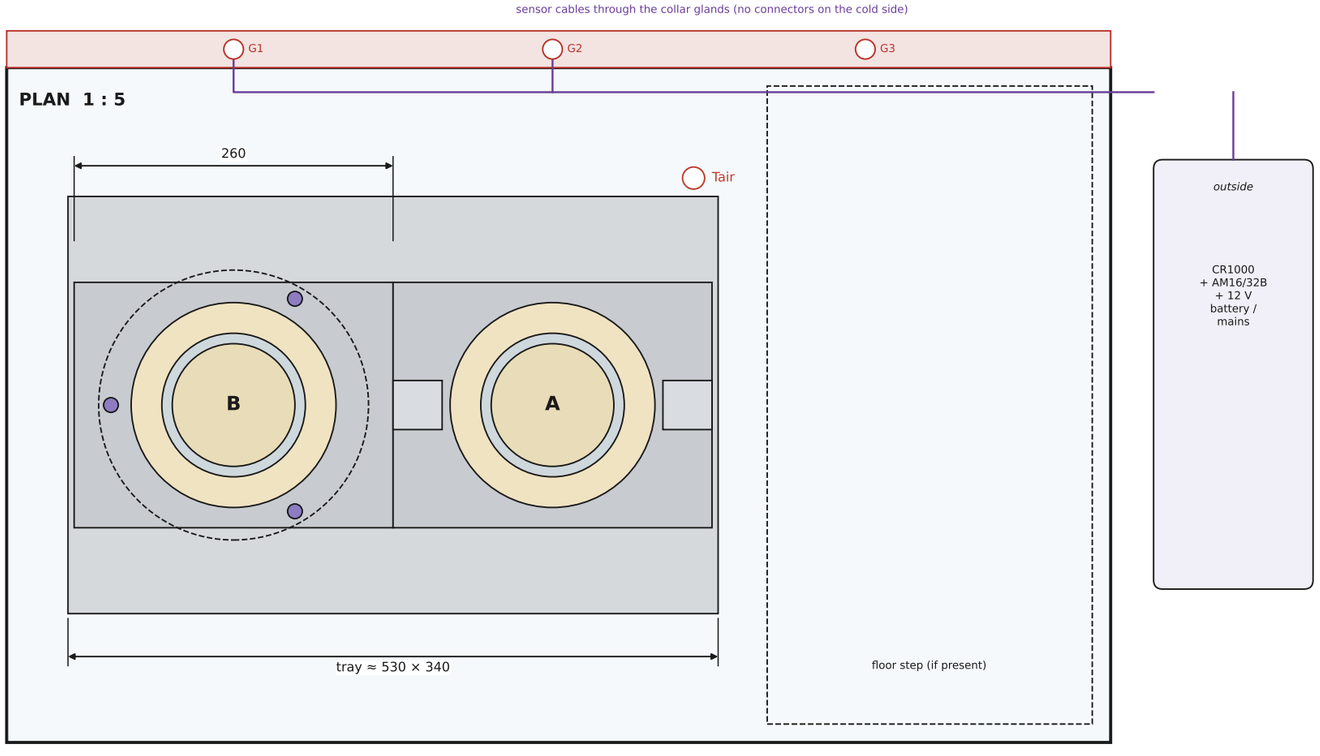
Resolution with the CR1000 (13-bit ADC, ±5000 mV range)
- LVDT ±10 mm (A): ≈ 1.3 µm; LVDT ±5 mm (B): ≈ 0.7 µm
  → layer strain over 25 mm ≈ $4 \cdot 10^{-5}$ (B)
- absolute linearity ±0.25 % FS → calibrate each LVDT with a micrometer
- 109SS: ±0.38 °C at 0 °C; ice-point offset check improves it

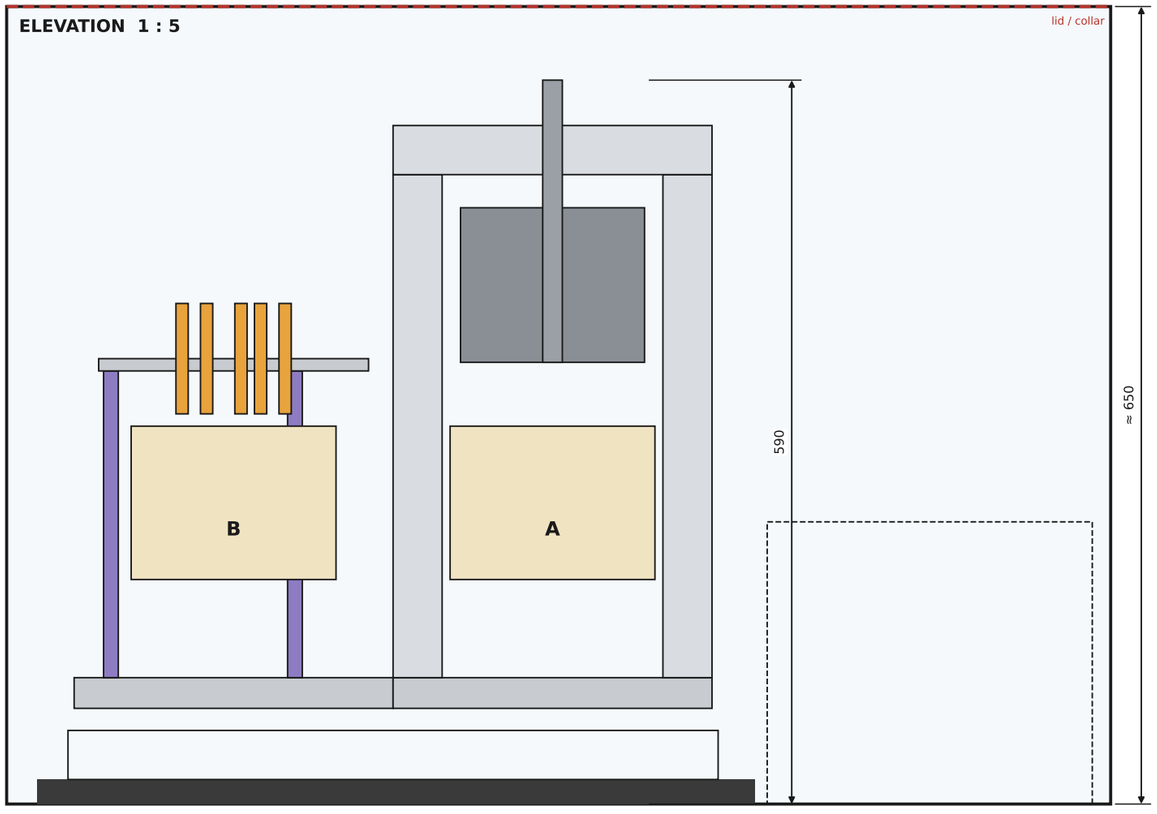
Notes
- Rig B uses LVDTs with a FREE (unguided) core – no spring force on the sample; the core rides on the anchor rod / plate rod. Check the free-core version and the core length with RDP before ordering [U].
- DC-LVDTs need their own ±15 V (or 24 V) supply, logger only reads ±5 V.
- CR1000 is discontinued (successor CR1000X / CR1000Xe, 24-bit); the existing one is fine for these resolutions.
- Logger outside the freezer (CR1000 −25…+50 °C).
- Point M: TDR, Pt100 and FBG sit 12 mm apart at one depth – Pt100 is the reference for the FBG and for T3; TDR gives unfrozen water vs. temperature.
- Pressure cells: calibrate in the same sand under dead weight (5–25 kPa) at +10 and −10 °C; correct zero(T) with the neighbouring thermistor.

both rigs on one CR1000 + AM16/32B; models from manufacturer data (see README §7)

Sensor selection and CR1000 wiring

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	–	drawing no.	**E7-V05**	sheet 5 / 10





Fit check

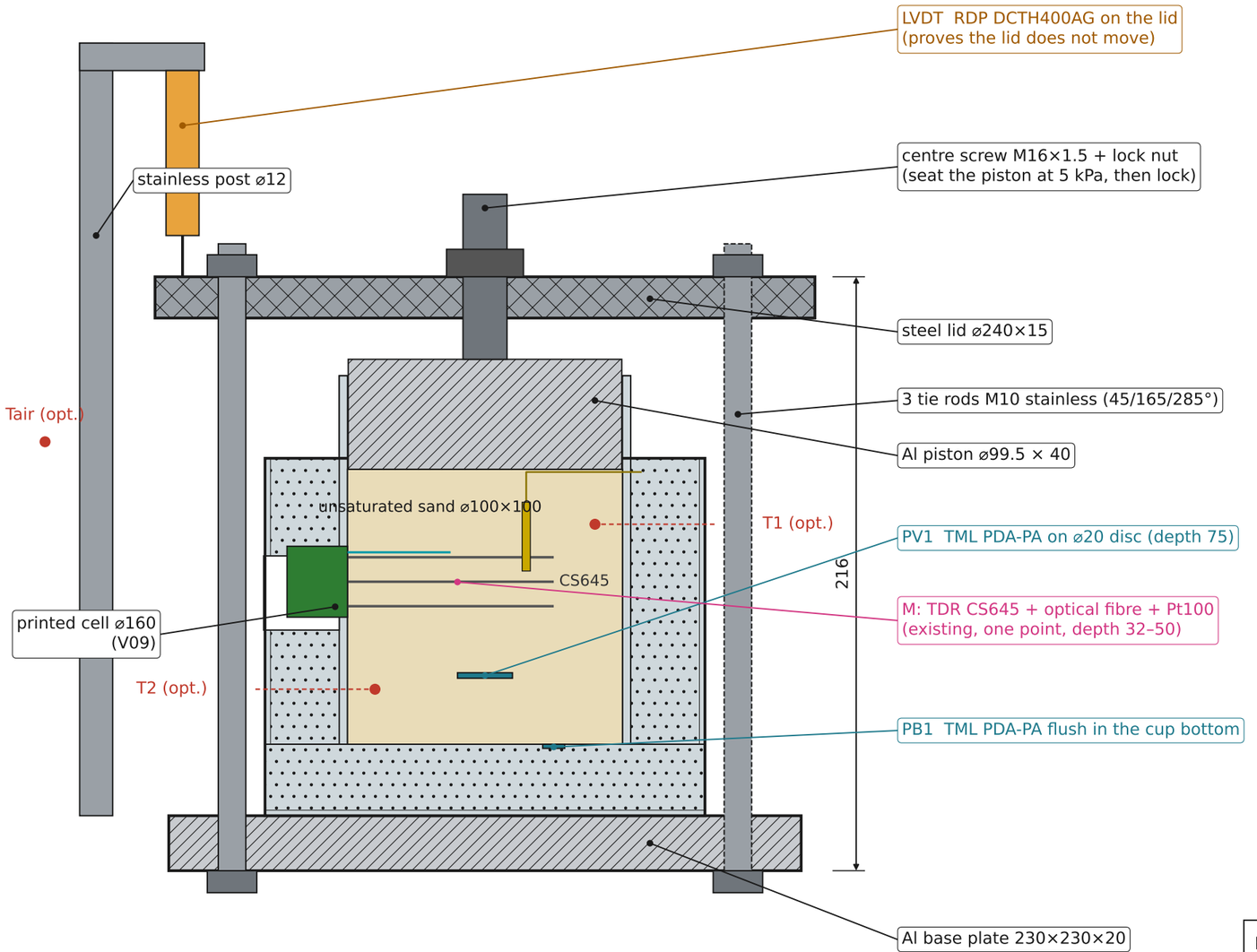
item	size	limit	status
base plates 2 × 260 × 200	520 × 200	tray ≈ 530 × 340	OK (tight)
rig A (weights)	20 kg steel discs ø150	nothing outside frame	OK
rig B (stand)	408 mm high	no load, no frame	OK
highest point	590 mm	inside ≈ 650	OK, measure
logger	outside, 0…+40 °C	CR1000 −25…+50 °C	keep warm & dry

Notes
- Both rigs share the freezer program (Set A) and the CR1000; each rig has its own Tair sensor at sample mid-height.
- Same sand, same printed cell, same thermistors in A and B – the only difference is the load (A ≤ 25 kPa dead weight, B none).
- Run the blank test with Al dummies in both rigs before the first sand run.
- Dimensions marked ≈ are estimates from the photos (E6) – please measure.

same freezer as E6 (inside ≈ 900 × 550 × 650, to be measured); logger outside

Both rigs in the chest freezer + CR1000

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 5**	drawing no.	**E7-V06**	sheet 6 / 10



To buy – type, quantity, price (EUR excl. VAT, ≈)

tag	type	pcs	EUR	CR1000
PV1, PB1	TML PDA-PA 500 kPa (ø6.5 × 1 mm) [U]	2	1 200–2 000	DIFF1, DIFF2 BrFull, VX1
LVDT	RDP DCTH400AG (±10 mm, spring)	1	581 *	SE9
+	DIN-rail power supply for the LVDT	1	40–80	–
T1, T2, Tair	Campbell 109SS-L (ø1.6 mm)	3	540–780	SE11–13, VX3

sensors ≈ 2 361–3 441 EUR mechanics ≈ 290–590 EUR → total ≈ 2 651–4 031 EUR
* price found at a dealer (Oct 2026, see README); others = estimates → ask for quotes.
Already available (no cost): TDR CS645, Pt100, optical fibre (point M), CR1000.

Existing sensors at point M

tag	type	connection
TDR	Campbell CS645, 3 rods ø3.2 × 75, pitch 8.9	TDR100/200, SDM
Pt100	Pt100, 37 mm (25 mm metal), vertical	4-wire
OF	optical fibre, tip 37.5 from the wall	own interrogator

Fixed lid (vypocty.md §9)
- piston seated by the centre screw at 5 kPa, locked → the sample cannot heave; freezing pressure is read by PV1 (in the sand) and PB1 (at the bottom)
- at 1 MPa: tie rods +49 µm, lid 60 µm → piston moves ≈ 0.1 mm (≈ 20 µm at 200 kPa)
- stainless tie rods and the inner stack shrink almost equally (Δ ≈ 12 µm per 20 K)

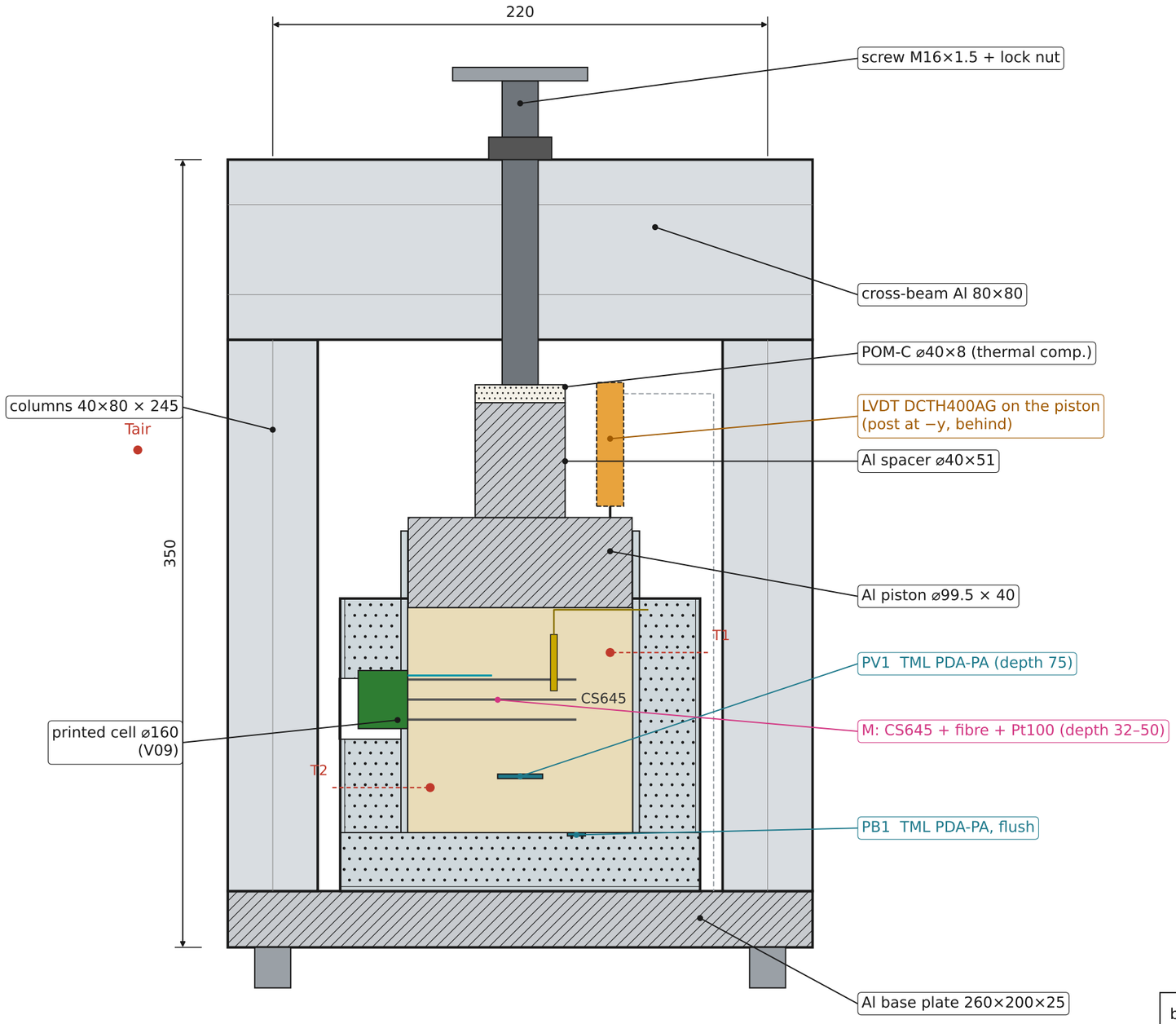
CR1000 wiring – no multiplexer
DIFF1 PV1, DIFF2 PB1 (BrFull, VX1) · SE9 LVDT · SE11–13 T1, T2, Tair (optional)
point M: per the existing devices (TDR on SDM C1–C3, temperature / fibre own reader)

Removed vs. variant A: frame, weights, carrier, ball, Invar posts, 2nd LVDT, 4 of 6 pressure cells, porous stones, pedestal, both multiplexers.

blue = pressure, red = temperature, magenta = point M

Variant A-min – fixed lid on 3 tie rods

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**≈ 1 : 2.2**	drawing no.	**E7-V07**	sheet 7 / 10



A-frame vs. A-min (tie rods)

	A-frame (this sheet)	A-min (E7-V07)
holds the piston	frame 40×80 + 80×80, screw	steel lid on 3 tie rods M10
piston movement at 1 MPa	≈ 0.08–0.15 mm	≈ 0.11 mm
thermal error (20 K)	≈ 0 with POM-C insert	≈ 12 µm
access to the sample	open sides, screw from top	lid must be removed
mechanics, EUR (≈)	≈ 375–760	≈ 330–670
sensors to buy, EUR (≈)	2 360–3 440 (same)	2 360–3 440

To buy (same sensors as A-min)

tag	type	pcs	EUR	CR1000
PV1, PB1	TML PDA-PA 500 kPa (ø6.5 × 1 mm) [U]	2	1 200–2 000	DIFF1, DIFF2 BrFull, VX1
LVDT	RDP DCTH400AG (±10 mm, spring)	1	581 *	SE9
+	DIN-rail power supply for the LVDT	1	40–80	–
T1, T2, Tair	Campbell 109SS-L (ø1.6 mm)	3	540–780	SE11–13, VX3

* price found at a dealer (Oct 2026); others estimated – ask for quotes.
Already available: TDR CS645 + reflectometer, Pt100, optical fibre (point M), CR1000.

How it works
- turn the screw until PV1 reads ≈ 5 kPa, lock the nut – the piston cannot move up;
- freezing pressure is read by PV1 (in the sand) and PB1 (bottom); LVDT proves the piston stays put (≈ 0.1 mm at 1 MPa, joints of the profiles dominate);
- Al columns shrink more than the sand stack – the POM-C insert expands the stack back (Δ ≈ 1 µm per 20 K), blank run with an Al dummy confirms it.

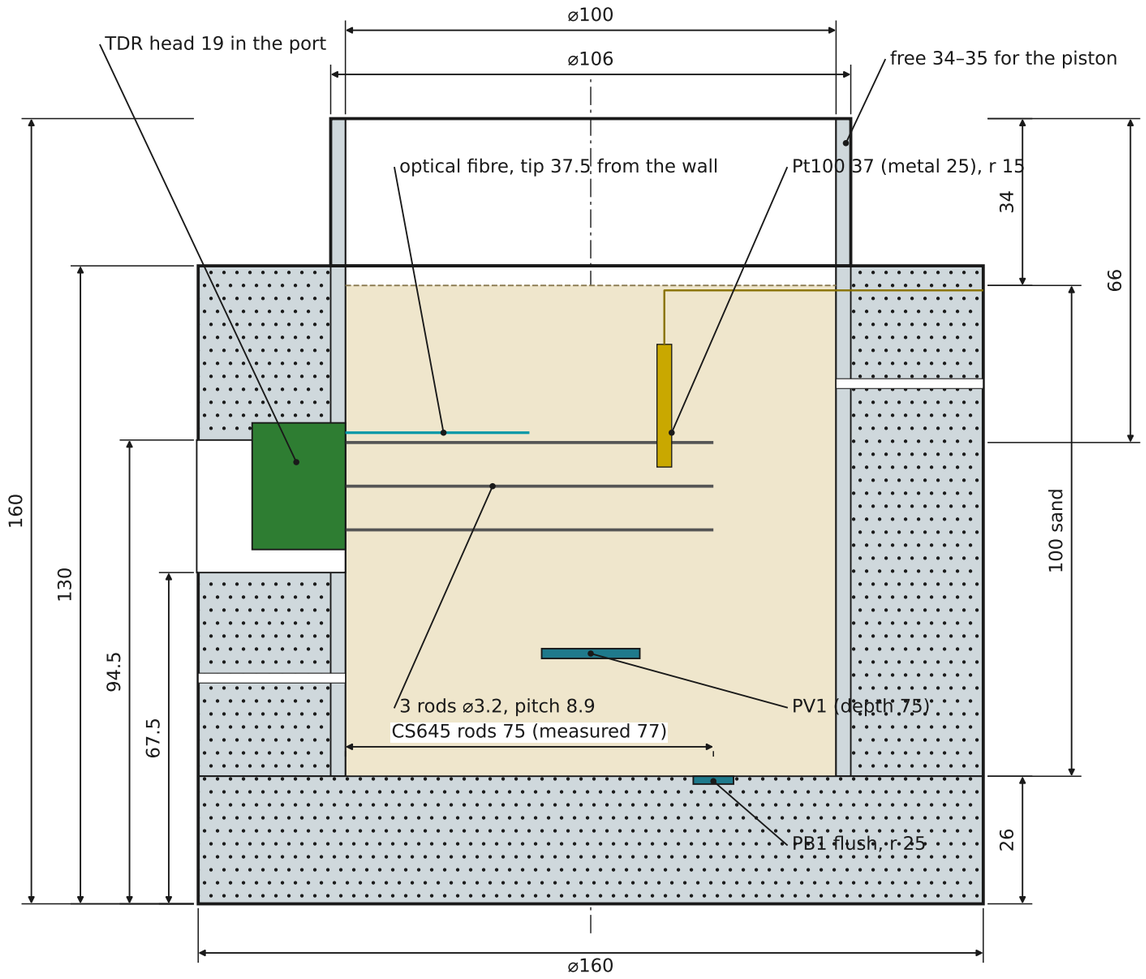
Details: vypocty.md §10.

blue = pressure, red = temperature, magenta = point M; dashed = behind

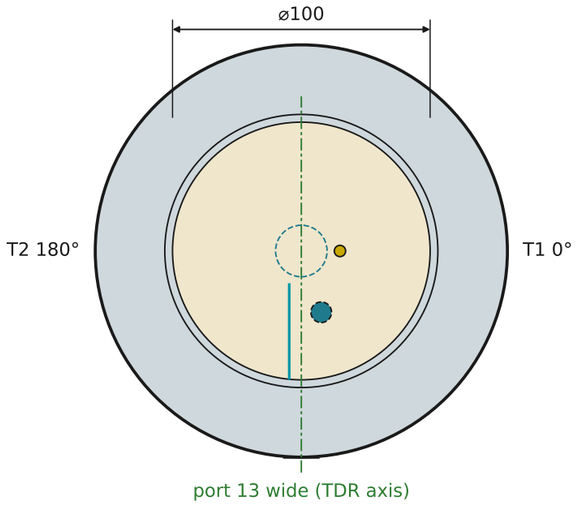
Variant A-frame – piston fixed by an Al frame

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**≈ 1 : 2.2**	drawing no.	**E7-V08**	sheet 8 / 10

SECTION A–A 1 : 1.25 (port and sensors revolved)



PLAN 1 : 2.4



Sensors at point M (existing)

	sensor	position
TDR	Campbell CS645	rods 75 (77 meas.), pitch 8.9; top rod 66 below the collar top
OF	optical fibre	horizontal at the top rod, tip 37.5 from the wall
Pt100	37 mm (25 metal)	vertical from the sand surface, 15 from the axis

Depths below the sand surface (z = 126): rods 32 / 41 / 50,
fibre ≈ 31, Pt100 metal 12–37 → all three sense the 30–50 mm layer.
Open (sketch 7.7.26): does the Pt100 measure at the tip or along its length?

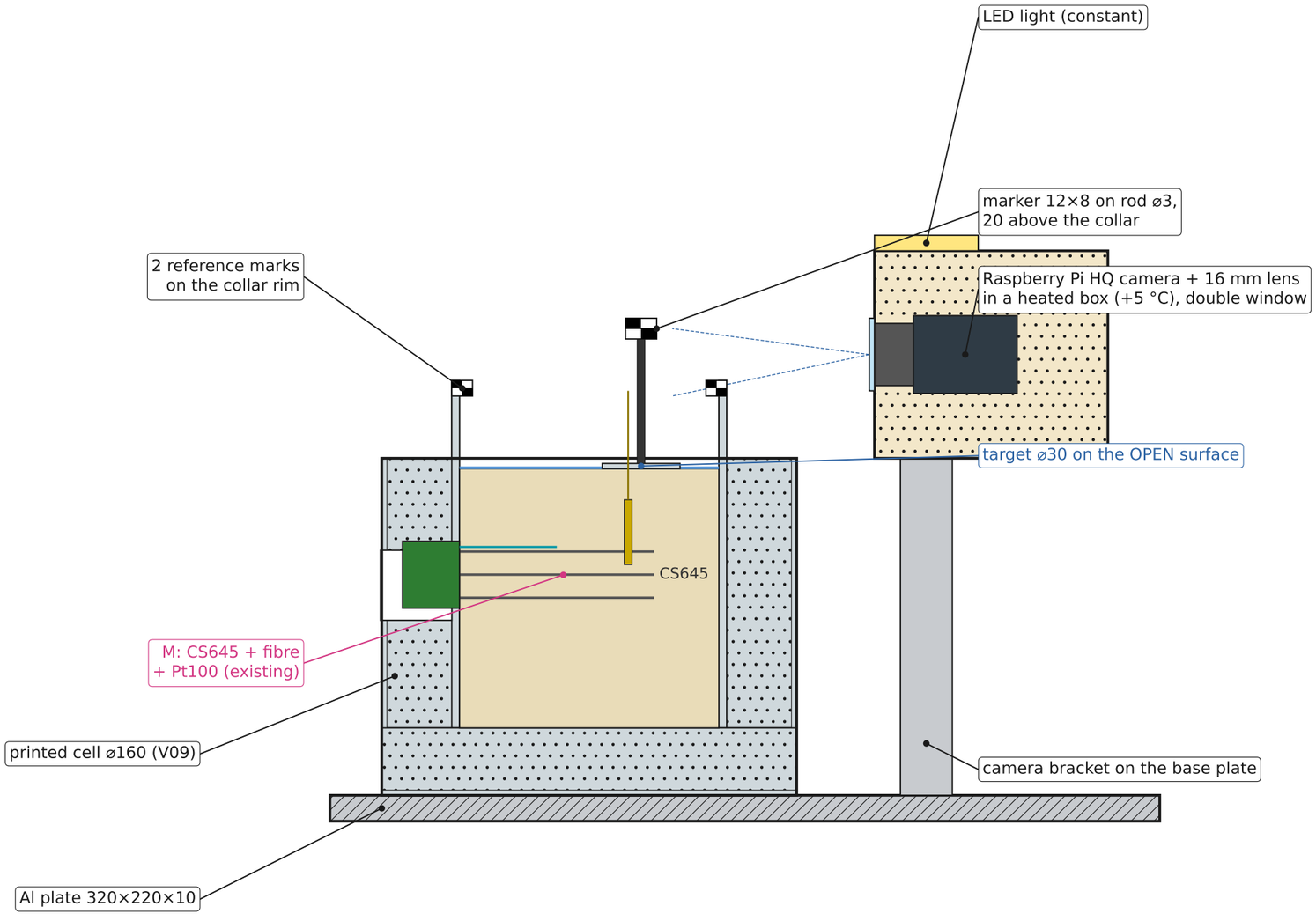
Cable routes (to print or drill)
- TDR head + coax and the fibre: through the side port 13 × 27
- Pt100 cable: groove under the piston, ⌀4 hole at z ≈ 125 (0°)
- PV1: ⌀3.5 hole at z = 51 (90°); PB1: groove in the cavity floor + ⌀3.5 hole through the jacket at 288°
- optional T1 (depth 20, 0°), T2 (depth 80, 180°): ⌀2 holes

Print (PrusaSlicer, MK4S): PETG(-CF), 0.2 mm, ≈ 600 g, 21 h
- jacket 10 % infill = insulation (replaces Kaiflex)
- recommended: 100 % infill under the cavity (z 0–26) – the bottom carries the full freezing pressure (10 % gyroid crushes at ≈ 1–2 MPa)
- wall ≈ 5 mm (≈ 8 perimeters) around the cavity: 3 mm alone sees ≈ 3 MPa hoop stress at 200 kPa, ≈ 17 MPa at 1 MPa
- epoxy coat + PTFE spray + silicone grease inside the cavity

per drawing M. Vrána 23.06.2026 + sensor sketch; z = 0 at the underside

Cell A – printed body ⌀160 (Column_one_body_version_1)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 1.25**	drawing no.	**E7-V09**	sheet 9 / 10



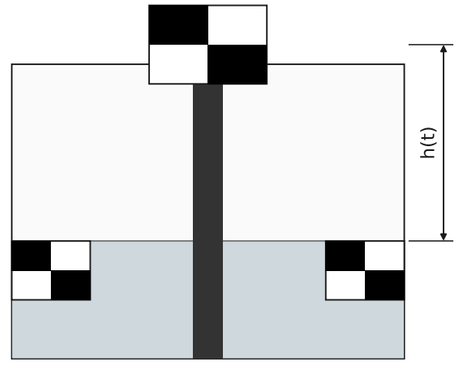
To buy (EUR excl. VAT, ≈)

tag	item	EUR
CAM	Raspberry Pi 5 (4 GB) + PSU + SD card	90–120
+	Raspberry Pi HQ Camera (IMX477, 12 MP)	55–70
+	lens 16 mm C-mount (+ spacer)	40–70
+	LED strip / ring 12 V, diffuser (constant light)	15–30
+	heated box: XPS 20, heater foil 5 W, thermostat, double-glass window	40–80
Tair	Campbell 109-L (optional)	150–220
mech.	Al plate 320 × 220 × 10	40–70
mech.	Al camera bracket (profile 20×20)	15–40
mech.	target PETG ⌀30 + rod ⌀3 + marker, 2 rim marks (printed)	2–10

Total ≈ 447–710 EUR (estimates). Already available: cell, CS645,
Pt100, optical fibre, CR1000. No LVDT, no Invar post.

Resolution ≈ 10 µm/px, sub-pixel ≈ 2–5 µm (vypocty.md §13)
Rim moves ≈ 120 µm per 20 K (PETG) → blank run with an Al dummy.
Photo every 10 min, clock synced (NTP) with the CR1000.

CAMERA VIEW (≈ 40 mm field) 1.6 : 1



heave(t) = h(t) − h(0); reference marks give scale (px→mm)
and cancel any movement of the camera

no lid, no LVDT, no Invar post: printed cell + point M + marker + camera

Variant B-open – open surface, displacement by camera

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**≈ 1 : 2.2**	drawing no.	**E7-V10**	sheet 10 / 10